Do Now · retrieval

Search engine indexing and PageRank

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is a hyperlink?

Q2

What is an index in retrieval?

Q3

What kind of graph models links?

Q4

BST root20, left10, right30; insert 25. Where is it placed?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — A crawler discovers pages by following links and sitemaps; an index records useful information about documents for searching.

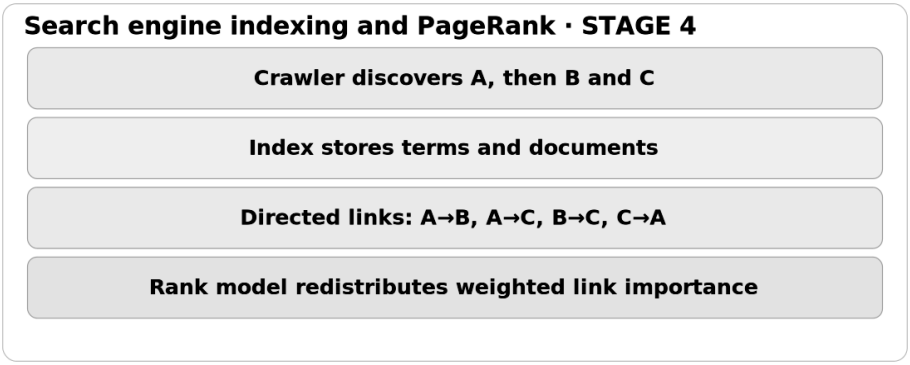
Core idea 2

Apply — PageRank models a random web surfer and passes importance through directed links. A link from a high-ranked page can carry more weight than one from a low-ranked page.

Core idea 3

Evaluate — PageRank uses damping and repeated iterations; raw incoming link counts alone are not the full algorithm. Search engines combine many ranking signals.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

crawler — Discovers/fetches documents.

inverted index — Maps terms to documents.

PageRank — Iterative link-based importance model.

damping — Probability factor in random-surfer model.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

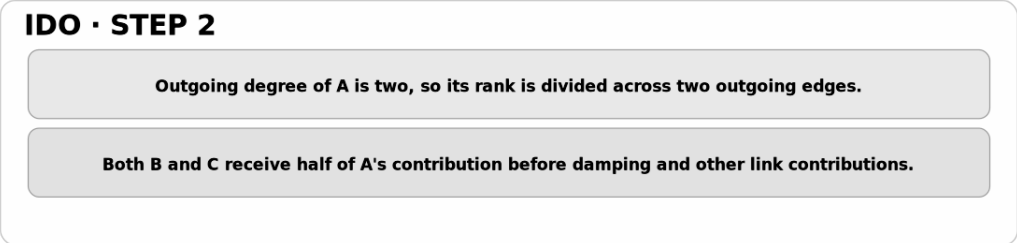
In graph A→B and A→C, how is A's rank contribution divided (ignoring damping)?

Follow the worked method

1. Predict first — Outgoing degree of A is two, so its rank is divided across two outgoing edges.

2. Explain — Both B and C receive half of A's contribution before damping and other link contributions.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first

The outgoing links split A's contribution evenly in the simplified model.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Distinguish crawling and indexing. [4]

Question 2

In directed link graph A→B,C and B→C, identify incoming links of C and A's share given rank 0.4, ignoring damping. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain why the number of inbound links alone is not an adequate PageRank calculation. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A page's PageRank is simply its number of incoming links.

Use key vocabulary from memory

1. Explain crawler in your own words.

2. Explain inverted index in your own words.

3. Explain PageRank in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.